# CS465 Machine Learning Project Instructions

## Project Overview

This project applies machine learning techniques to a real-world predictive or classification analytics problem and requires students to present their work in the form of an IEEE-style research paper. Students will design, implement, evaluate, and critically analyze a complete machine learning pipeline, emphasizing experimental rigor, reproducibility, and scholarly communication.

### Objectives

- Apply data preprocessing and feature engineering techniques in real-world datasets.
- Design and evaluate multiple machine learning models.
- Analyze and interpret experimental results.
- Communicate technical findings using IEEE research paper standards.

### Paper Structure Requirements

**Submission Format:** The primary submission must be an IEEE conference-style paper (6 pages) or IEEE Access-style journal paper (10 -12 pages), it should include the following sections:

- Abstract
- Introduction
- Related Work (Literature review)
- Dataset Exploration: Preprocessing, Data Analysis, and Feature Engineering
- Methodology
- Experimental Results
- Discussion
- Conclusion and Future Work
- References

### Deliverables

- IEEE-format paper (word document).
- Github repo containing the full ML pipeline.
- Dataset or data access instructions.
- 5-minute presentation summarizing the research.

### Evaluation Criteria

- Research clarity and problem formulation.
- Correct application of ML techniques.
- Quality of experimental evaluation and analysis.
- Critical discussion of limitations and insights.
- Writing quality and IEEE formatting compliance.
- Reproducibility of results.

### Academic Integrity and AI Usage

Generative AI tools may be used to assist with coding or writing. However, students must fully understand, explain, and justify all submitted work. Any use of AI tools must not replace original analysis or critical thinking.

## Tasks and Milestones

### Week 3: Problem Definition and Dataset Selection

- Define a clear **research problem** and prediction or classification target.
- Justify **dataset** choice and relevance to the problem domain.
- Identify **state-of-the-art or commonly used benchmarks** relevant to your task.
    - Identify **baseline approaches** (5–10 recent peer-reviewed papers related to your topic published in the last 2–3 years).
- Draft the **Introduction** section, including:
    - Motivation
    - Problem formulation
    - Initial statement of intended contributions.

**Deliverable:**

- Approved dataset
- 1–2 page Introduction draft
- List of baseline models to be used for benchmarking

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### Week 5: Related Work and Benchmark Identification

- Review the **5–10 recent peer-reviewed papers related to your topic (last 2–3 years)**.
- For each paper, extract:
    - Task definition
    - Models, data, and features used
    - Evaluation metrics and experimental setup
- Explicitly state **limitations or gaps** in existing methods that your project will address.

**Deliverable:**

- Related Work section draft
- Summary table comparing prior methods, datasets, and metrics
- Defined benchmark metrics to be used in your experiments

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### Week 7: Dataset Characterization, Preprocessing, and EDA

- Provide a formal dataset description consistent with research papers:
    - Types of data (images, text, audio, numerical), size, feature types, class distribution.
- Perform Preprocessing Steps: Cleaning, normalization, and handling of missing values.
- Annotation/Labeling: Details on how labels were generated, especially for supervised learning.
- Perform EDA to:
    - Identify biases, imbalances, or noises
    - Support modeling and evaluation decisions
- Relate observations to **issues reported in prior studies**.

**Deliverable:**

- Dataset and Preprocessing section draft
- EDA figures and statistical summaries
- Justification of evaluation protocol (train, test, and cross-validation split)

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### Week 10: Feature Engineering and Baseline Reproduction

- Engineer features informed by:
  - EDA findings
  - Techniques used in prior literature
- Implement **baseline models** reported in related work when feasible.
- Ensure features and baselines enable **fair comparison**.

**Deliverable:**

- Feature Engineering section draft
- Description of baseline implementations
- Initial baseline performance results

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### Week 15: Methodology, Benchmarking, and Experimental Evaluation

- Describe full experimental setup:
  - Model architectures
  - Hyperparameter tuning strategy
- Train **multiple ML models** and compare them against:
  - Simple baselines
  - Reproduced literature baselines (when possible)
- Report results using **standardized metrics** aligned with prior work.
- Perform comparative analysis:
  - Quantitative performance comparison
  - Error analysis

**Deliverable:**

- Methodology section
- Experimental Results section (tables and figures)
- Benchmark comparison results
- Discussion section

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### Week 16: Full Paper Draft (IEEE Format)

- Integrate all sections into a coherent IEEE paper.
- Include:
  - Clear comparison with existing work
  - Discussion of performance gaps or improvements
- Ensure all figures, tables, and claims are supported by experiments.
- Write introduction, conclusion, and abstract (following this order).

**Deliverable:**

- Complete draft paper in IEEE format
- Internal consistency check between paper, code, and results

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### Week 17: Final IEEE Paper Submission

- Incorporate instructor feedback.
- Ensure reproducibility and clarity.

**Deliverable:**

- Final IEEE-format paper (PDF)
- GitHub repository with finalized code and documentation

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### Week 19: 5-minute Presentation

- Present the work as a research contribution:
    - Problem and motivation
    - Methodology (System Overview Diagram)
    - Benchmarks and baselines (table)
    - Comparative results
    - Key insights and limitations

**Deliverable:**

- 5-minute presentation + 10-minute Q&A

## Grade Distribution

| Milestone | Grade |
|---|---|
| Problem Definition and Dataset Selection | 20 |
| Related Work and Benchmark Identification | 20 |
| Dataset Characterization, Preprocessing, and EDA | 20 |
| Feature Engineering and Baseline Reproduction | 20 |
| Methodology, Benchmarking, and Experimental Evaluation | 30 |
| Full Paper (IEEE Format) | 10 |
| Presentation | 30 |
| Total | 15% |

## Suggested topics:

### 1) Trustworthy and Explainable AI for High-Stakes Prediction

Focuses on making machine learning decisions transparent, reliable, and interpretable in sensitive domains such as healthcare and finance.

Good for XAI + calibration + uncertainty + robustness.

- **Sepsis early warning (clinical tabular)** Kaggle
- **ICU outcome / mortality prediction (EHR-style)** Kaggle
- **Credit risk benchmark (financial risk, strong for explainability + fairness)** Kaggle
- **Cervical cancer risk factors (tabular medical prediction, common in XAI papers)** Kaggle

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### 2) Multimodal Learning for Human Behavior and Engagement Analysis

Studies how combining visual, textual, and sensor data improves the prediction of human behavior and engagement.

- **RISE-MICCAI Fair Federated AI competition (medical imaging classification, can be used for multimodal extensions and fairness benchmarking)** Kaggle
- **OpenStreetMap (buildings/roads/POIs for geo context to fuse with mobility or pollution)** OpenStreetMap Download
- **NYC TLC Trip Record Data (mobility signals, can be fused with weather/air quality)** New York City Government New York City Taxi and Limousine Commission
- **OpenAQ (air quality, can be fused with mobility, weather, events)** OpenAQ+1

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### 3) Privacy-Preserving ML and Federated Learning

Investigates decentralized learning methods that protect data privacy while maintaining predictive performance..

- **RISE-MICCAI Fair Federated AI competition (2024)** Kaggle
- **"Datasets for Federated Learning" (collection)** Kaggle
- **Federated Health Records dataset** Kaggle
  **OpenAQ (privacy-friendly public sensors, useful for FL across cities)** Open Data on AWS OpenAQ

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### 4) Bias, Fairness, and Equity in Real-World ML

Analyzes and mitigates demographic bias in machine learning models to ensure fair and equitable outcomes.

- **Realistic Loan Approval Dataset | US & Canada** Kaggle

- **City Lifestyle Segmentation dataset** Kaggle
- **Credit Card Fraud Detection Dataset** Kaggle
- **Credit Risk benchmark dataset (also suitable for fairness constraints + explainability)** Kaggle

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## 5) Time-Series Forecasting Under Distribution Shift

Examines how predictive models adapt to changing data patterns in real-world temporal data.

- **Time Series Forecasts (benchmark collection)** Kaggle
- **OpenAQ (hourly updates, multi-country)** Open Data on AWS
- **AI Traffic Optimization System** Kaggle
- **Hourly Crypto & Stocks Market Data Kaggle**
- **Time Series in IOT (Internet of Things) Kaggle**
- **Delhi 5-minute electricity demand (high-frequency forecasting)** Kaggle

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## 6) Graph ML for Real-World Relational Data

Applies graph-based learning to model complex relationships among entities in networks and structured data..

- **KG20C Scholarly Knowledge Graph (papers/citations style KG)** Kaggle
- **Graph ML Contest (link prediction setup on Kaggle)** Kaggle
- **OpenStreetMap (a massive real-world graph for routing, POI graphs, spatial KGs)** OpenStreetMap Download
- **OpenAQ (build station graphs (spatial/temporal edges) for spatio-temporal GNNs)** Open Data on AWS+1

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## 7) AI for Sustainable Cities and Smart Infrastructure

Uses machine learning to support urban sustainability through traffic, energy, and environmental analytics.

- **NYC TLC Trip Record Data (official, continuously updated)** New York City Government+1
- **OpenAQ (air quality, global)** Open Data on AWS+1
- **Smart Mobility Traffic Dataset (Kaggle)** Kaggle **Or Kaggle**
- **Smart Traffic Signal Management System** Kaggle

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## 8) End-to-End ML Pipelines with Reproducibility and Benchmarking

Focuses on building and evaluating complete, reproducible machine learning pipelines using standardized benchmarks.

- **OpenML task collections (benchmark suites, curated)** OpenML+1
- **OpenML-CTR23 regression benchmark suite (paper + suite on OpenML)** OpenReview+1
- **AutoML Benchmark framework (tabular benchmarks, reproducible comparisons)** openml.github.io
- **NYC TLC Trip Record Data (industry-scale benchmark for pipelines)** New York City Government+1